

Phase 1 Documentation

AI-Powered Supply Chain Risk & Response Copilot (POC)

1. Executive Summary

Organizations today have access to large amounts of supply chain visibility data, including shipment tracking, supplier updates, logistics events, financial indicators, and market intelligence. However, these systems primarily tell users **what happened**, not **what business impact it creates** or **what action should be taken**.

The objective of this initiative is to build an **Agentic AI-powered Supply Chain Risk & Response Copilot** that can:

1. Detect supply chain disruptions.
2. Quantify potential financial impact.
3. Recommend mitigation actions.

The long-term vision discussed in [Supply Chain end to end visibility and impact .docx](#) is to create an intelligent supply chain decision-support platform capable of translating disruptions into measurable business outcomes and actionable recommendations.

2. Why Are We Building This?

Current Challenge

Most supply chain systems provide:

- Shipment status
- Delay notifications
- Supplier alerts
- Risk signals

However, decision makers still need to manually answer:

- What products are affected?
- How much revenue is at risk?
- What will be the impact on margins?
- Should we expedite shipments?
- Should we switch suppliers?
- What is the optimal response strategy?

This causes:

- Slow decision-making
 - Reactive operations
 - Manual analysis
 - Increased business risk
-

Business Goal

Instead of:

1 A supplier in Japan is affected by an earthquake.

The platform should provide:

1 Revenue at Risk: \$2.5M
2 Margin Impact: \$350K
3 Recommended Action:
4 Expedite critical shipments
5 Expected Recovery:
6 2 weeks

This transforms visibility into actionable decision intelligence.

3. Phase 1 Scope

Phase 1 focuses on proving the concept using a small but meaningful set of AI agents.

The objective is to demonstrate end-to-end flow:

1 Risk Detected
↓
2 Impact Calculated
↓
3 Recommendation Generated

4. Phase 1 Agent Architecture

Agent 1: Risk / Sensing Agent

Objective

Detect supply chain disruptions and convert them into structured risk events.

Scope

Monitor:

- Earthquakes
- Port congestion
- Supplier disruptions
- Shipment delays
- Financial risk indicators

Technology Options

Option A - Open APIs

Tavily API

Sources:

- News sites
- Public web content
- Industry updates

Option B - Flat Files

Sample file:

```
1 risk_events.csv
```

Containing:

```
1 Event Type
2 Location
3 Severity
4 Supplier
5 Date
```

Output

```
{
  "event": "Earthquake",
  "location": "Japan",
  "severity": "High"
}
```

Agent 2: Financial Impact Agent

Objective

Determine the financial consequences of a disruption.

Scope

Calculate:

- Revenue at Risk
- Margin Loss
- Penalty Costs
- Expedite Costs

Technology Options

Option A - Yahoo Finance API

Enrichment:

- Supplier market performance
- Financial strength indicators
- Market risk signals

Option B - Flat Files

Example:

```
1  supplier_revenue.csv
```

Containing:

```
1  Supplier
2  Revenue Impact
3  Margin %
4  Penalty Cost
5  Expedite Cost
```

Output

```
1  {
2    "revenueRisk":2500000,
3    "marginLoss":350000,
4    "penaltyCost":200000,
5    "expediteCost":150000
6  }
```

Agent 3: Recommendation Agent

Objective

Act as the AI decision-maker.

Technology

Groq API

Supported models:

- Llama
- DeepSeek
- Mixtral

Responsibilities

Evaluate:

- Risk severity
- Revenue impact
- Cost of mitigation
- Business trade-offs

Generate:

- Recommended actions
- Justification
- Recovery expectations

Output

Recommended Action:

Expedite shipments

Expected Benefit:

Protect \$2.35M revenue

Recovery Timeline:

2 weeks

5. Demo Workflow

Step 1

Risk Agent identifies:

1 Earthquake detected near Supplier A

Step 2

Financial Agent calculates:

- | |
|--|
| <ol style="list-style-type: none">1 Revenue Risk = \$2.5M2 Penalty Exposure = \$200K3 Margin Loss = \$350K |
|--|

Step 3

Recommendation Agent generates:

- | |
|---|
| <ol style="list-style-type: none">1 Recommended Action:2 Expedite critical inventory3 Justification:4 Mitigation cost is lower than projected revenue loss.5 Outcome:6 Revenue Protected = \$2.35M |
|---|

6. Data Strategy

Option 1: Flat File Driven POC

Advantages:

- Rapid development
- No dependency on enterprise systems
- Easier demonstrations

Data Sources:

- | |
|--|
| <ol style="list-style-type: none">1 Risk Events.csv2 Supplier Financials.csv3 Revenue Impact.csv |
|--|

Recommended for Initial Demo

Option 2: Open API Driven POC

Advantages:

- Real-time data
- Demonstrates live intelligence
- Better executive showcase

Data Sources:

- | |
|--|
| <ol style="list-style-type: none">1 Tavily API |
|--|

- | | |
|---|-------------------|
| 2 | Yahoo Finance API |
| 3 | Groq API |

Recommended for Innovation Showcase

7. Deliverables

Phase 1 will deliver:

Functional POC

- Risk Detection Agent
- Financial Impact Agent
- Recommendation Agent

User Interface

Simple dashboard displaying:

- Detected risks
- Financial impact
- AI recommendations

Demonstration Scenarios

- Earthquake disruption
 - Supplier financial downgrade
 - Shipment delay
 - Port congestion
-

8. Success Criteria

The Phase 1 POC will be considered successful if it can:

- ✓ Detect a disruption
- ✓ Calculate business impact
- ✓ Generate an AI-recommended response
- ✓ Demonstrate end-to-end decision support
- ✓ Support both flat-file and API-based execution

9. What Comes Next (Phase 2)

After validating the proof of concept, additional agents will be introduced.

Exposure Agent

Determines:

- Impacted suppliers
- Impacted plants
- Impacted materials
- Impacted products

Inventory Agent

Determines:

- Stockout risks
- Safety stock violations
- Inventory exposure

Customer Impact Agent

Calculates:

- OTIF impact
- Customer impact
- Sales order exposure

Scenario Simulation Agent

Simulates:

- Do Nothing
- Expedite
- Alternate Supplier
- Inventory Reallocation

This expands the platform from disruption awareness to business simulation.

10. Long-Term Vision (Phase 3)

Build a complete **Multi-Agent Supply Chain Decision Intelligence Platform**.

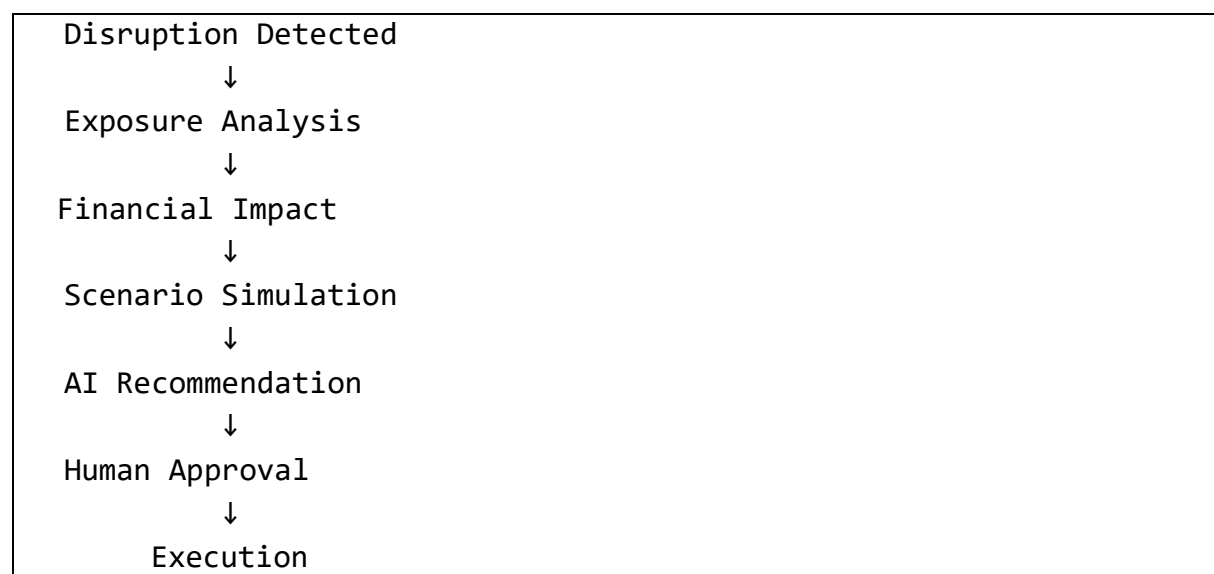
Enterprise Integrations

- SAP ERP
- SAP IBP
- Ariba
- TMS
- MES

Advanced Agents

- Exposure Agent
- Capacity Agent
- Inventory Agent
- Customer Impact Agent
- Financial Agent
- Scenario Simulator
- Recommendation Agent

End-State Vision



The final platform becomes an **AI Supply Chain Control Tower**, providing real-time disruption visibility, business impact quantification, and AI-driven response recommendations, aligning with the strategic vision discussed by [Selwyn DSouza](#) and [Ravin Angara](#).